Decentralized Cyber-Physical Stabilization of Coupled Kuramoto Oscillators via Emergent MARL CSMA/CD Protocols

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Abstract

AI data centers executing distributed training workloads impose periodic, spatially correlated active-power disturbances on transmission grids. We demonstrate that uncoordinated, synchronous software cycles act as a low-frequency periodic driving force that drives the grid into a period-doubling bifurcation cascade and localized blackouts. We present a decentralized Multi-Agent Reinforcement Learning (MARL) framework that discovers an emergent CSMA/CD protocol at the grid interface boundary, restoring global phase-locking and stability without centralized scheduling.

1 Introduction and Problem Formulation

Conventional energy management systems evaluate substations independently, failing to extract the cross-substation statistical structures that characterize synchronized load episodes. When high-performance deep learning clusters execute recurring batches (e.g., approximately 50 seconds per batch on DGX hardware), they generate localized active-power surges. To model this interaction, we append the aggregate data center load $P_{\mathrm{AI},i}(t)$ directly into the second-order Kuramoto swing equations for a grid node i:

$$m_i\ddot{\theta}_i + d_i\dot{\theta}_i = P_i - \sum_{j=1}^{N} K_{ij}\sin(\theta_i - \theta_j) - P_{\mathrm{AI},i}(t) \tag{1}$$

where m_i represents the rotational turbine inertia, d_i represents the natural damping, and K_{ij} defines the admittance coupling strength of the high-voltage transmission lines. The bounds of global phase-locking under these non-linear conditions follow the fundamental alignment proofs established by *Chopra & Spong (2009)*.

2 Multi-Scale Control Models and High-Fidelity EMT Grounding

We implement a dual-resolution physical framework to model grid safety. Macroscopic stability is governed by second-order dynamics to isolate inter-area phase angle bifurcation cascades. Micro-level, sub-millisecond electromagnetic transient (EMT) numerical simulations are executed across a modified IEEE 39-bus New England testbed system using the native, open-source Julia framework

`PowerSimulationsDynamics.jl` documented by *Lara et al. (2023)* to model non-linear inverter-based resource (IBR) Phase-Locked Loop (PLL) voltage tracking dynamics.

When a data center drops high-amplitude periodic training steps on an IBR network, the low-frequency forcing signal interacts with high-speed internal current controller switching loops to induce severe control-loop interaction chaos, forcing the local bus tracking voltage $V_i(t)$ to undergo catastrophic transient phase-slips and localized blackouts. These sub-millisecond variations are captured as extensional exemplars sampled at a resolution of $\Delta t = 0.5$ ms:

$$\boldsymbol{\xi}_i(t) = [V_i(t),\, \delta_i(t),\, P_{\mathrm{IBR},i}(t),\, Q_{\mathrm{IBR},i}(t)] \in \mathbb{R}^4 \tag{2}$$

As reinterpreted within the contemporary multi-time-scale grid hierarchies of *Dall'Anese, Simpson-Porco, Dörfler, et al. (2026)*, integrating feedback optimization directly with first-principles physical dynamics enables new decentralized coordination regimes for highly heterogeneous resources.

3 Hierarchical Pattern Recognition and Scalable Parameter Estimation

To achieve automated system identification without centralized computational overhead, we construct an advanced hierarchy of structural patterns. Time-based input-output trajectories are harvested from synchronous generators and inverter-based resources under both nominal states and phase noise conditions. To recover hidden physical constants and unmapped dynamics from partial measurements, we implement a scalable bi-level optimization scheme fusing the Prediction Error Method (PEM) with a Moving Horizon Estimation (MHE) state estimator as formulated by *Zhou & Köhler (2024)* and accelerated via parametric tuning paradigms by *Simpson et al. (2024)*.

Let the non-linear parameterized grid device state space model be represented by the discrete-time mapping:

$$x_{k+1} = f_d(x_k, u_k, w_k; \theta) \tag{3}$$

$$y_k = Cx_k + F\eta_k \tag{4}$$

where $x_k \in \mathbb{R}^{n_x}$ is the latent state, $u_k \in \mathbb{R}^{n_u}$ is the control input vector, $y_k \in \mathbb{R}^{n_y}$ is the measured telemetry output, and w_k, η_k represent mutually independent process and measurement noise structures. The unknown parameters $\theta \in \mathbb{R}^{n_\theta}$ are recovered by segmenting input-output data into a dataset $\mathcal{D} = \{u^{(i)}_{0:N-1}, \tilde{y}^{(i)}_{0:N}\}_{i=1}^{M}$ of M overlapping time-series windows of length N. To guarantee absolute physical interpretability alongside flexible residual mapping, the algebraic observation function tracks a physics-neural gray-box model structure:

$$h_{\mathrm{gray}}(x, u; \theta_{\mathrm{gray}}) := h_{\mathrm{phys}}(x, u; \theta_{\mathrm{phys}}) + h_{\mathrm{NN}}(x, u; \theta_{\mathrm{NN}}) \tag{5}$$

where h_{phys} defines the explicit physics-based turbine constants and h_{NN} acts as a neural network residual tracker mapping anomalous load spikes.

By eliminating parameter-wide sensitivity matrices, hypergradient evaluation realizes up to a 215× speedup, allowing the real-time parameter estimates to be outputted instantly as discrete features and values. These values form positive and negative structural exemplars ($\mathcal{E}$) to retrain our localized decision forest memory matrices, establishing certifiable models to verify the correctness of any active inter-area network configuration.

4 Throughput Derivation and Data Center Parameter Estimation

Following the network routing principles of *Sherman & Gable (1983)*, the emergent CSMA/CD protocol evaluates carrier availability using a normalized propagation delay parameter $a = \tau_{\text{prop}}/T_{\text{trans}}$. Given that electricity travels through copper busbars and high-voltage cabling at approximately $v = 0.67c \approx 2 \times 10^8$ m/s, a massive 1000-meter campus cable run maps to a propagation delay of $\tau_{\text{prop}} = 5$ μs. Bound against our sub-millisecond EMT control sample window of $T_{\text{trans}} = 500$ μs, the data center power network parameter is estimated exactly at $a = 0.01$.

The absolute non-persistent CSMA channel throughput (S) as a function of offered traffic load (G) is governed by the structural formula pioneer-analyzed by *Kleinrock (1975)*:

$$S = \frac{G \cdot e^{-aG}}{G \cdot (1 + 2a) + e^{-aG} - (1 - e^{-aG})} \tag{6}$$

When $a = 0.01$, the network realizes an exceptionally efficient operational curve, enabling our decentralized MARL agents to maximize collective rewards and execute randomized delay gates before transient voltage drops breach perimeters. When conflicts emerge, agents initialize a preference-ordered dialogue game (Scounter $\succ$ Sarg $\succ$ Sconcede $\succ$ Sclose) based on the algebraic principles of *Black, Hunter & Pan (2009)* to safely align load profiles over narrow-band protocol channels.

5 Simulation Performance Readouts and Trajectory Plots

We record the performance of both physical grid architectures. The text-notational plot below documents the absolute maximum macroscopic Kuramoto frequency deviation ($|\dot{\theta}_i|$) across a 40-step simulation execution continuum, tracking the transition from open-loop bifurcation to phase-locked stabilization under active MARL access gating:

```
STEP | FREQ DEVIATION (w) | GRAPHICAL TRAJECTORY (w vs. Time)
-----------------------------------------------------------------------
   0 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   1 | 0.6500             | ######################## [PHASE SLIP]
   2 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   3 | 0.3500             | #############
   4 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   5 | 0.6500             | ######################## [PHASE SLIP]
   6 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   7 | 0.3500             | #############
   8 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
   9 | 0.6500             | ######################## [PHASE SLIP]
  10 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
  11 | 0.9500             | ##################################... [BIFURCATION]
  12 | 1.1000             | ######################################### [DROOP]
  13 | 0.6500             | ######################## [PHASE SLIP]
  14 | 0.5000             | ###############---> [OPEN-LOOP CHAOS]
  15 | 0.9500             | ##################################... [BIFURCATION]
  16 | 0.9500             | ################################### [MARL INTERV]
  17 | 0.8234             | ##############################
  18 | 0.7102             | ############################
```

```
 19 | 0.6189            | ########################
 20 | 0.5342            | ###################### [DAMPING ENGAGED]
 21 | 0.4611            | ################## [PHASE-LOCKED STATE]
 22 | 0.4012            | ################ [PHASE-LOCKED]
 23 | 0.3510            | ############ [PHASE-LOCKED]
 24 | 0.3090            | ########### [PHASE-LOCKED]
 25 | 0.2711            | #========= [STABLE ATTAINABLE SPACE V']
 26 | 0.2394            | ######### [PHASE-LOCKED]
 27 | 0.2105            | ######## [PHASE-LOCKED]
 28 | 0.1876            | ####### [PHASE-LOCKED]
 29 | 0.1654            | ###### [PHASE-LOCKED]
 30 | 0.1492            | ##### [PHASE-LOCKED]
 31 | 0.1321            | ##### [PHASE-LOCKED]
 32 | 0.1198            | #### [PHASE-LOCKED]
 33 | 0.1085            | #### [PHASE-LOCKED]
 34 | 0.0984            | ### [PHASE-LOCKED]
 35 | 0.0892            | ### [PHASE-LOCKED]
 36 | 0.0811            | ### [PHASE-LOCKED]
 37 | 0.0734            | ## [PHASE-LOCKED]
 38 | 0.0672            | ## [PHASE-LOCKED]
 39 | 0.0610            | ## [PHASE-LOCKED]
-------------------------------------------------------------------
```

The data tables below contrast the high-fidelity sub-millisecond Julia-to-Python EMT simulation results across the modified IEEE 39-bus inverter topology:

5.1 Phase 1: Open-Loop Repetitive Forcing (Unstable IBR Control Chaos)

```
FORCING WORKLOAD  | VOLTAGE MAGNITUDE (p.u.) | FREQUENCY SWING (Hz) | STATUS
-------------------------------------------------------------------------------
 14.50 MW         | 0.9448                   | -0.6580              | CRITICAL CHAOS
-------------------------------------------------------------------------------
```

5.2 Phase 2: Closed-Loop Active MARL Access Gating Stabilization

```
FORCING WORKLOAD  | VOLTAGE MAGNITUDE (p.u.) | FREQUENCY SWING (Hz) | STATUS
-------------------------------------------------------------------------------
 14.50 MW         | 0.9985                   | -0.0015              | PHASE-LOCKED
-------------------------------------------------------------------------------
```

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